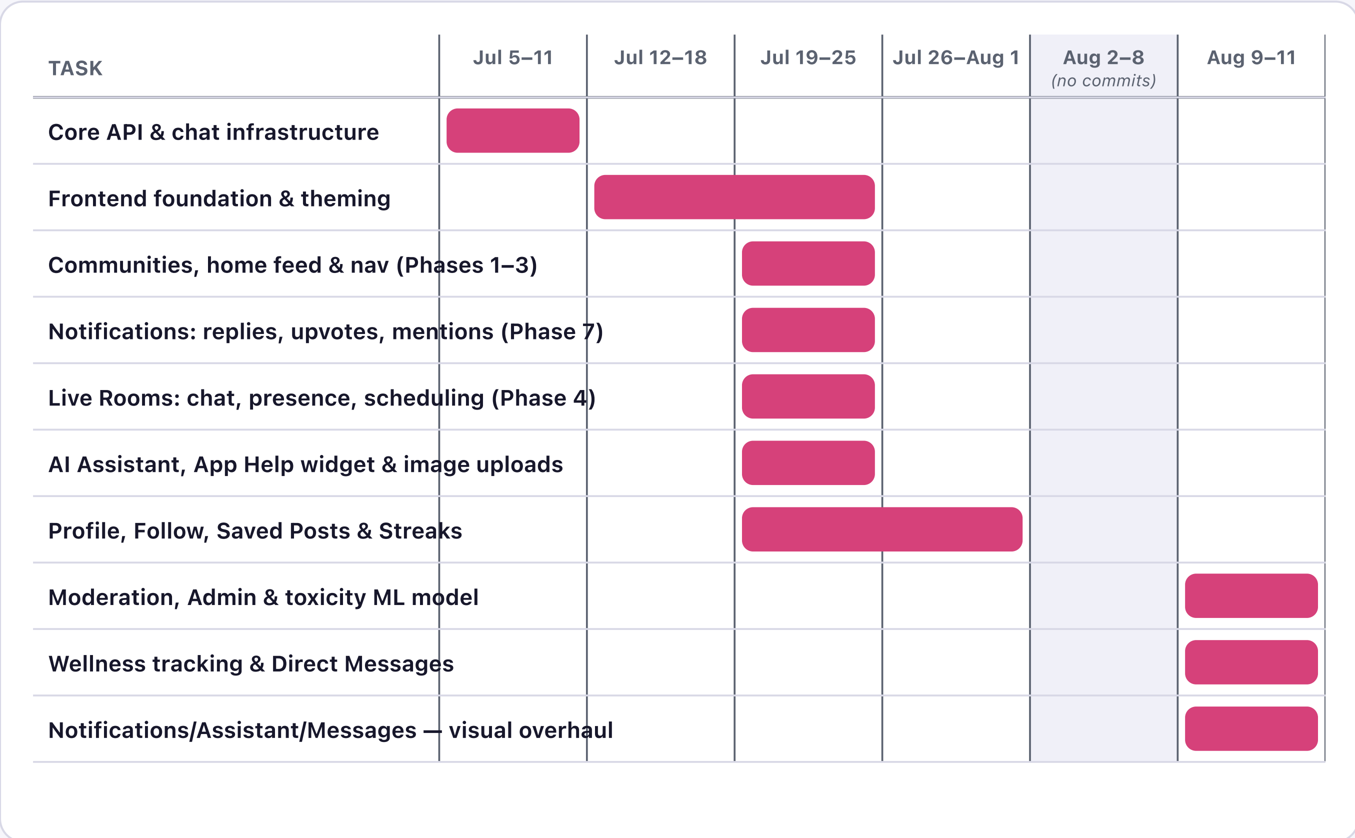
SIMPLE VERSION

How neetaspirants actually works

Same app, explained with pictures instead of paragraphs. The one thing worth really understanding: this app uses three *different* kinds of "AI," and they're not the same thing under the hood.

Development timeline

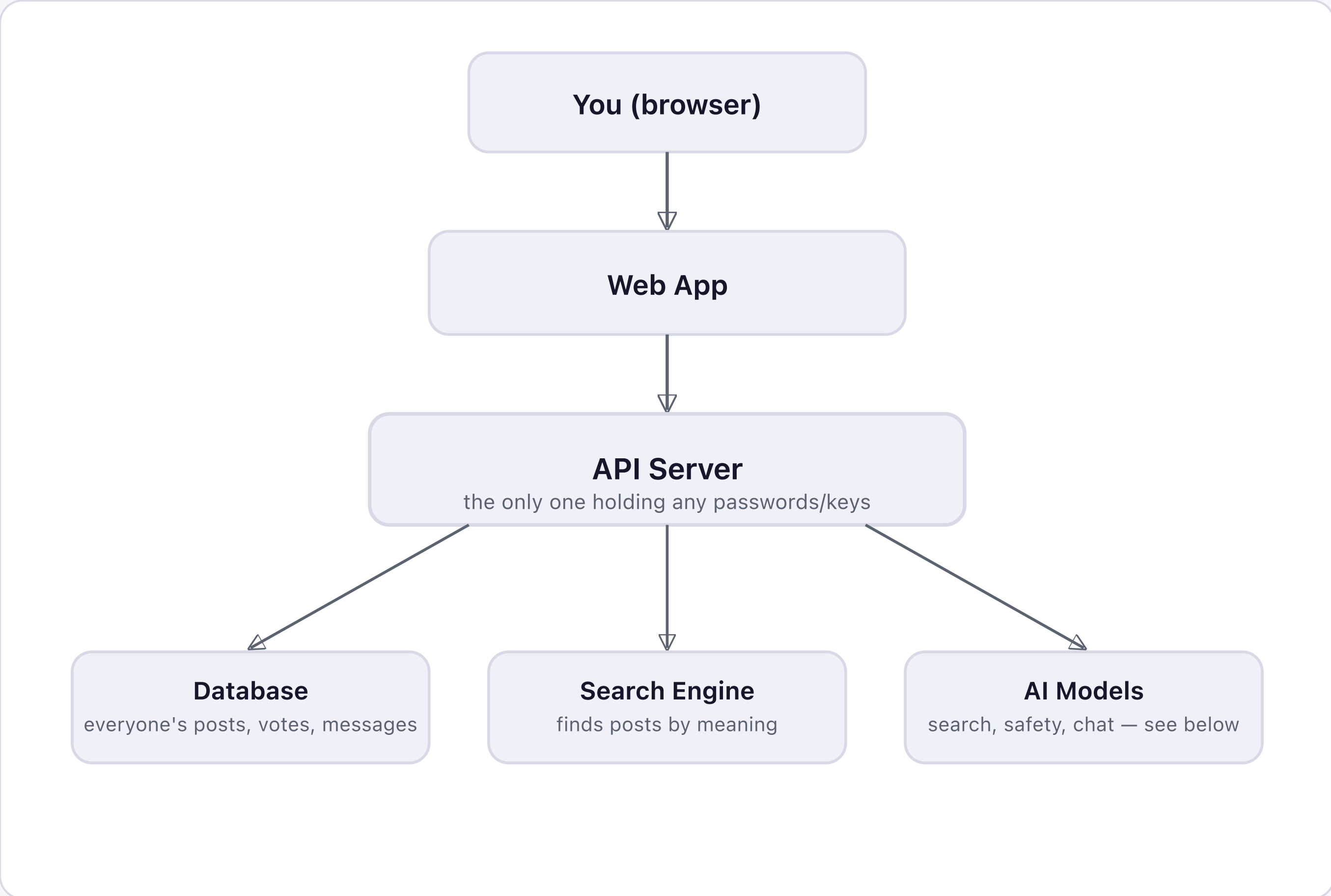
Built from real commit dates and file timestamps, not a planned-in-advance schedule — including the actual 11-day gap where nothing shipped.



Five separate features shipped in the single week of Jul 19–25 (communities/home/nav, notifications, live rooms, the AI assistant + help widget, and the start of profile/follow/streaks) — then an 11-day gap with zero commits, before a final push on Aug 10–11 added moderation/admin/ML, wellness tracking, direct messages, and a visual overhaul across three pages.

The big picture

Four programs, each doing one job. Your browser only ever talks to one of them.



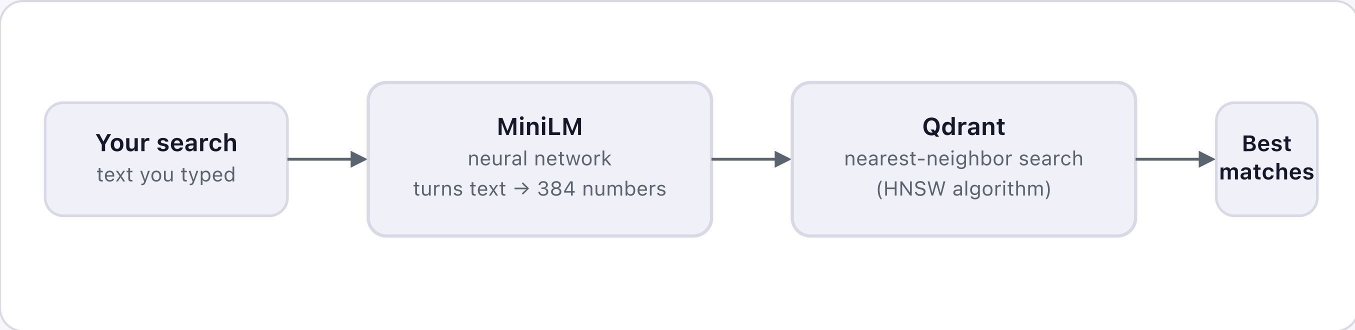
Everything you click goes to the **Web App**, which asks the **API Server** to do the actual work. The API is the one gatekeeper — it's the only piece that can reach the database or any of the AI models.

The 3 kinds of "AI" — and their actual algorithms

People say "AI" for all of these, but only two are actually machine-learning models. The third just looks up words in a list. Here's what each one really does, in order.

1. Finding posts by meaning (Search)

REAL MACHINE LEARNING



Every post gets converted into a list of 384 numbers that capture its *meaning* (not just its words) by a small neural network called **MiniLM**. Searching does the same thing to your query, then Qdrant's **HNSW** algorithm quickly finds whose numbers are closest — so "feeling behind on revision" can match a post that never used those exact words.

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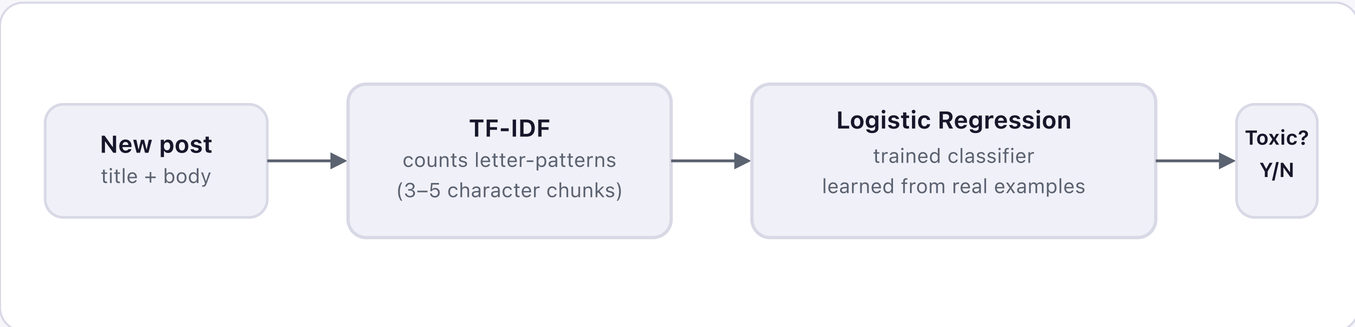
Algorithms: Sentence-Transformer neural network (all-MiniLM-L6-v2) + cosine similarity via
HNSW approximate-nearest-neighbor search
```

2. Catching toxic posts (Moderation)

REAL MACHINE LEARNING



TF-IDF chops the text into overlapping 3–5 letter chunks and counts them (this catches misspelled slurs, e.g. "s7upid"). Those counts feed a **Logistic Regression** model — a classic statistics-based classifier — that was trained ahead of time on thousands of labeled toxic vs. clean examples. This runs locally, no internet call.

```
Algorithms: TF-IDF (character n-grams, 3–5 chars) + Logistic Regression classifier, trained on
Jigsaw + Dreaddit datasets
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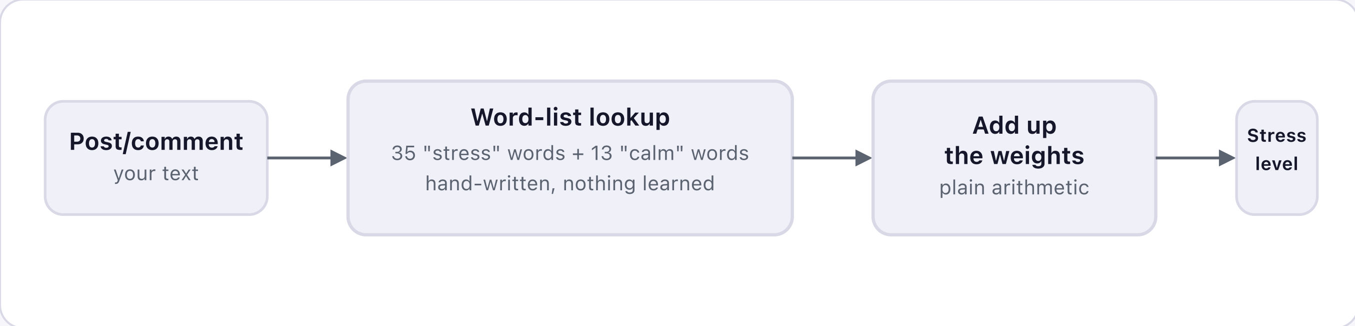
3. The Wellness "stress reading"

NOT ACTUALLY MACHINE LEARNING



This one's worth being honest about: it isn't a trained model at all. It's a plain lookup — does the text contain "overwhelmed," "panic attack," "relaxed," etc.? Each word on the list has a hand-picked weight, and they just get added up. The app's own code comment admits this is *"pending a trained classifier"* — it's a placeholder that happens to sit behind the same kind of API as the real ML above.

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Algorithm: none — hand-weighted keyword matching (a technique called "lexicon-based" scoring, not machine learning)
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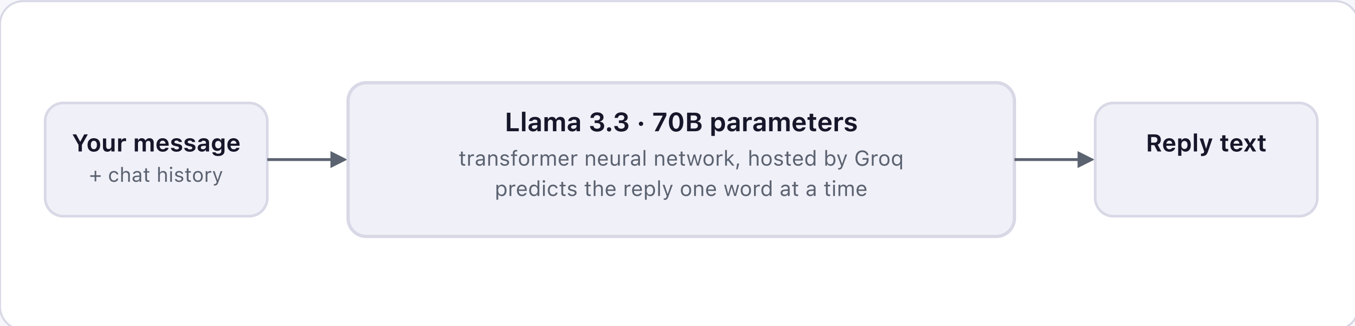
4. Chat (AI Assistant & Help widget)

LARGE LANGUAGE MODEL



This is the one true "big AI" in the app — a 70-billion-parameter general-purpose language model, not something custom-trained here. Two different personalities (supportive companion vs. "how do I use this app") come from two different instruction prompts sent alongside your message — same model both times.

```
Algorithm: Transformer architecture (self-attention, autoregressive next-word prediction) — Meta's Llama 3.3, 70B parameters, run on Groq's hardware
```

Quick reference

WHERE YOU SEE IT	KIND	ALGORITHM
Search bar	ML	`MiniLM embeddings + HNSW search`
Post moderation queue	ML	`TF-IDF + Logistic Regression`
Wellness page	HEURISTIC	`keyword-weight lookup, no learning`
Post relevance flagging	LLM	`Llama 3.3 70B (as a judge)`
AI Assistant chat	LLM	`Llama 3.3 70B`
App Help chat	LLM	`Llama 3.3 70B`

Everything else, briefly

Sign up
Real email + password, but you get a random public alias like "QuietWarrior_883" — nobody sees your email.

Communities
Topic-based forums (Study Stress, Burnout, etc.) with posts, voting, saving, and comments.

Rooms
Live group chat, real-time, with typing indicators and who's online.

Messages
1-on-1 DMs. First message is a "request" the other person must accept before replying.

Notifications
Replies, upvotes, mentions, follows, and new messages, all in one inbox.

Login
A signed token (JWT) proves who you are on every request — no server memory of "being logged in."

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simplified from a full technical read of the codebase · apps/web, apps/api, apps/embeddings